A+ Hands-On Technical Project Report

Mac Computer Diagnostics and IT Support Lab

Project ID: APLUS-LAB-001

Prepared by: Aftabuddin Muzamel Kalakani

Computer: MacBook Pro (16-inch, 2023)

Processor: Apple M2 Pro

Memory: 16 GB

Operating System: macOS Sequoia 15.6.1

Report Type: Technical Diagnostics and Troubleshooting

Project Status: Diagnostic testing and technical report completed; project presentation in progress.

Table of Contents

A+ Hands-On Technical Project Report **1**

Mac Computer Diagnostics and IT Support Lab *1*

1. Executive Summary **3**

2. Project Objectives **3**

3. Computer Environment **3**

4. Diagnostic Methodology **3**

5. Diagnostic Test Results **5**

Test 001 – Identify Mac Hardware and Operating System *5*

Test 002 – Inspect Storage Usage *5*

Test 003 – Identify Installed Memory *5*

Test 004 – Verify macOS Version and Build *5*

Test 005 – Identify Network Interfaces *5*

Test 006 – Check Wi-Fi IPv4 Address *5*

Test 007 – Identify Default Gateway *6*

Test 008 – Test Local Gateway Connectivity *6*

Test 009 – Test Public IP Connectivity *6*

Test 010 – Test DNS Name Resolution *6*

Test 011 – Review Network Service Order *6*

Test 012 – Inspect DNS Configuration *7*

6. Key Findings **8**

7. Limitations **8**

8. Conclusion **8**

9. Future Improvements **8**

1. Executive Summary

This project was completed to gain practical experience with computer diagnostics, operating system inspection, and basic network troubleshooting using macOS.

The project involved 12 diagnostic tests using built-in macOS tools and Terminal commands. The tests focused on identifying hardware information, reviewing storage and memory, examining network configuration, testing connectivity, and inspecting DNS settings.

The results demonstrated successful connectivity to the local network gateway and a public IP address, as well as successful DNS resolution for google.com.

2. Project Objectives

The main objectives were to:

- Identify computer hardware and operating system information.
- Review installed memory and storage usage.
- Identify network interfaces and IP configuration.
- Identify the default gateway.
- Test local and external network connectivity.
- Verify DNS name resolution.
- Interpret diagnostic results.
- Practice technical documentation and troubleshooting skills.

3. Computer Environment

Component	Recorded Information
Computer	MacBook Pro (16-inch, 2023)
Processor	Apple M2 Pro
Installed Memory	16 GB
Operating System	macOS Sequoia 15.6.1
Build Number	24G90
Storage Used	286.6 GB
Total Displayed Storage	994.66 GB

4. Diagnostic Methodology

The investigation used built-in macOS utilities and read-only Terminal commands.

The diagnostic process followed these steps:

1. Collect system and hardware information.

2. Inspect storage and installed memory.
3. Identify network interfaces and IP configuration.
4. Identify the default gateway.
5. Test network connectivity.
6. Inspect DNS resolution and configuration.
7. Record and interpret the results.

No hardware components were removed or replaced, and no destructive system changes were performed.

5. Diagnostic Test Results

Test 001 – Identify Mac Hardware and Operating System

Tool: Apple menu → About This Mac

Result: MacBook Pro (16-inch, 2023), Apple M2 Pro processor, macOS Sequoia 15.6.1.

Finding: The computer model, processor, and operating system were successfully identified.

Test 002 – Inspect Storage Usage

Tool: System Settings → General → Storage

Result: 286.6 GB used out of 994.66 GB displayed storage.

Finding: Approximately 708.06 GB remained available based on the displayed figures.

Test 003 – Identify Installed Memory

Tool: Apple menu → About This Mac

Result: 16 GB unified memory.

Finding: The installed memory capacity was successfully identified.

Test 004 – Verify macOS Version and Build

Tool: Terminal – `sw_vers`

Result: macOS Sequoia 15.6.1, Build 24G90.

Finding: The operating system version and build number were successfully verified.

Test 005 – Identify Network Interfaces

Tool: Terminal – `networksetup -listallhardwareports`

Result: The Mac listed Wi-Fi and several other network interfaces. Wi-Fi used interface en0.

Finding: The network interfaces and their corresponding hardware ports were identified.

Test 006 – Check Wi-Fi IPv4 Address

Tool: Terminal – `ipconfig getifaddr en0`

Result: Private IP address assigned.

Finding: The Wi-Fi interface had a private IPv4 address assigned.

Test 007 – Identify Default Gateway

Tool: Terminal – `route -n get default`

Result: Gateway Local default gateway identified, interface en0.

Finding: The Mac's default route used the Wi-Fi interface and the identified gateway.

Test 008 – Test Local Gateway Connectivity

Tool: Terminal – `ping -c 4` Local default gateway identified

Result: Four packets transmitted, four received, zero packet loss. Average response time: 9.840 ms.

Finding: The Mac successfully communicated with its local network gateway during the test.

Test 009 – Test Public IP Connectivity

Tool: Terminal – `ping -c 4 8.8.8.8`

Result: Four packets transmitted, four received, zero packet loss. Average response time: 26.206 ms.

Finding: The Mac successfully reached the public IP address during the test.

Test 010 – Test DNS Name Resolution

Tool: Terminal – `nslookup google.com`

Result: The DNS lookup successfully returned multiple IP addresses for google.com.

Finding: DNS name resolution was successful during the test.

Test 011 – Review Network Service Order

Tool: Terminal – `networksetup -listnetworkserviceorder`

Result: The Mac listed USB Ethernet, Thunderbolt Bridge, Wi-Fi, and iPhone USB services. A separate default-route check showed Wi-Fi (en0) was being used.

Finding: Network services were identified, and Wi-Fi was confirmed as the interface used by the default route.

Test 012 – Inspect DNS Configuration

Tool: Terminal – `scutil --dns`

Result: The Wi-Fi resolver listed DNS servers 68.105.28.11, 68.105.29.11, and 68.105.28.12.

Finding: DNS configuration was successfully inspected. The resolver reported the network as reachable.

6. Key Findings

The diagnostic investigation produced the following findings:

- The Mac's hardware and operating system information were successfully identified.
- Storage usage and installed memory were reviewed.
- The Wi-Fi network interface and private IPv4 configuration were identified.
- The default gateway was identified.
- Local gateway connectivity was successful, with zero packet loss.
- Public IP connectivity was successful, with zero packet loss.
- DNS name resolution for google.com was successful.
- DNS server configuration was inspected.

Overall, the tests demonstrated that the inspected system and network functions were operating successfully during the diagnostic checks.

7. Limitations

This project was limited to basic, read-only diagnostics using built-in macOS tools.

The tests do not establish that every hardware component is free from defects or that every network service is functioning correctly.

The results represent the system's condition at the time of testing. No hardware stress tests, component replacements, or destructive troubleshooting procedures were performed.

8. Conclusion

This project provided practical experience with computer system inspection, network configuration, connectivity testing, DNS troubleshooting, and technical documentation.

By completing 12 diagnostic tests, I strengthened my understanding of several CompTIA A+ concepts and gained hands-on experience using macOS Terminal commands and system utilities.

The project demonstrates my ability to follow a structured diagnostic process, collect technical information, interpret basic results, and document findings professionally.

9. Future Improvements

Potential improvements for future versions of this project include:

- Add screenshots of selected diagnostic commands and results.
- Perform additional troubleshooting scenarios.
- Explore macOS system logs and hardware diagnostic tools.

- Expand the project with Windows troubleshooting exercises.
- Add more practical CompTIA A+ hardware and operating system labs.